Transformation and Modeling Logic

1. Data Transformation

In this project, I used Power Query Editor in Power BI to clean and organize the e-commerce data before analysing it. The project contains three CSV files: List of Orders, Order Details, and Sales Target. First, I checked the data types and column quality to identify any missing values or errors. I limited the List of Orders table to 500 rows as required by the assignment and formatted the customer names properly. I also combined the City and State columns into a single Location column to make location-based filtering easier.

Next, I created a Profit Margin column to understand the profit earned compared to the sales amount and a Profit Status column to identify whether a record resulted in a profit, loss, or break-even. I then merged the List of Orders and Order Details tables using the common Order ID column to create a new table called Orders Data. I also created separate summary tables to calculate the order count by category, average profit by category, and total amount by sub-category. The Sales Target data was prepared for monthly and category-wise analysis.

2. Data Modeling Logic

After transforming the data, I worked on the relationships between the tables in Power BI Model view. I used common columns, such as Order ID, to connect related tables where appropriate. These relationships help organize the data and make it easier to analyse information from different tables together. While setting up the model, I encountered an ambiguous relationship path error because multiple connections existed between tables. I reviewed the relationships and removed an unnecessary connection to resolve the issue.

The main purpose of the transformation and modeling process was to make the data cleaner, easier to understand, and more useful for analysis. The prepared tables can help compare category-wise profit, sub-category sales amounts, order counts, and sales targets. Overall, this project helped me understand how Power Query is used to prepare data and how relationships in Power BI help organize data for meaningful analysis.